

StayAnalytics — Preliminary Report

StayAnalytics — Preliminary Project Proposal

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Convert this Markdown file to **prelim.pdf** before submission (e.g. `pandoc prelim.md -o prelim.pdf` or print-to-PDF from a viewer).

1. Project title

StayAnalytics — Cross-Platform OTT Movie & Series Analytics

2. Problem and domain

Streaming-video subscribers and casual analysts increasingly juggle multiple OTT platforms (Netflix, Prime Video, Hotstar, ...). Each platform maintains its own opaque ranking (“trending now”, “top 10”) that is not directly comparable across services, and most platforms do not expose a *community* rating distinct from third-party sources like IMDb. As a result:

- Discovering high-quality content requires hopping between apps.
- It is hard to compare, say, Netflix’s drama catalog against Prime Video’s, or to see which platform has the highest average user rating in a given genre.
- Personal “to-watch” lists are siloed inside each app.

This project addresses those gaps with a single web application backed by a unified relational database of titles, platforms, and community reviews.

3. Solution

StayAnalytics is a Spring Boot + Thymeleaf web app where any user can:

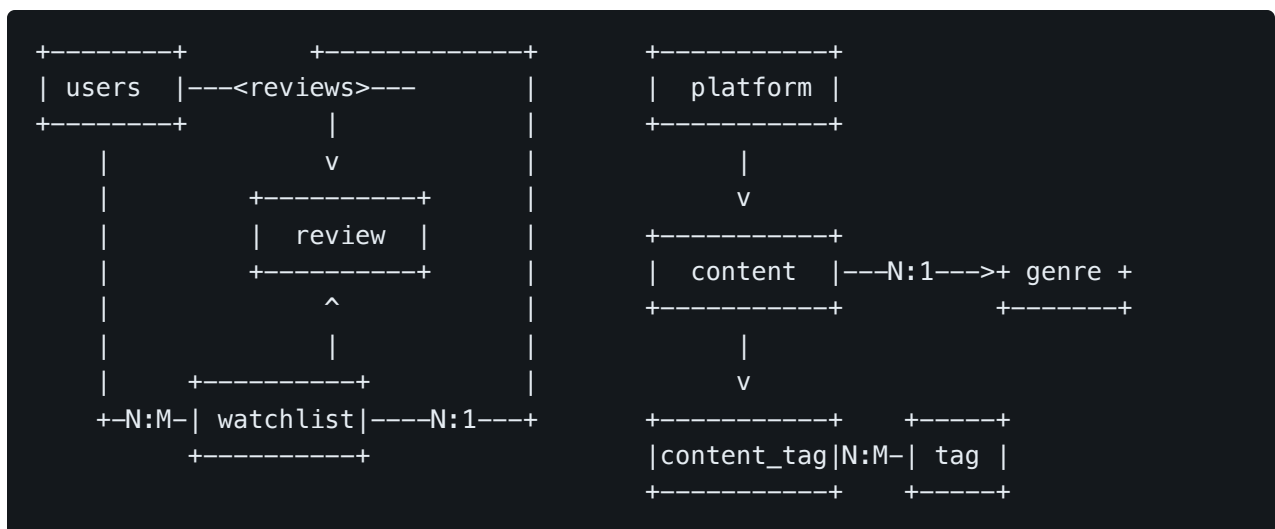
- **Browse** trending titles across platforms (`/home`).
- **Search/filter** by title, genre, platform, year, and type (`/search`).
- **View detail** for any title — IMDb metadata, our community rating, tag-based “similar titles” recommendations, and the review feed (`/movies/{id}`).

Authenticated users additionally:

- **Sign up / log in** with passwords stored as BCrypt hashes.
- **Maintain a personal watchlist** with status `PLAN_TO_WATCH` / `WATCHING` / `WATCHED` (`/watchlist`).
- **Rate and review** any title; one review per (user, title) pair with upsert semantics.

Anyone can open a cross-platform **analytics dashboard** that surfaces aggregate insights: average rating per platform, genre×platform breakdown, and top community-rated titles (`/analytics`). An admin role unlocks an additional `/admin` summary page.

4. Preliminary ER diagram



Entity sets (≥ 4 required): 1. `users` — accounts with role USER/ADMIN. 2. `platform` — Netflix, Prime Video, Hotstar. 3. `genre` — Action, Comedy, Crime, Drama, Fantasy, Sci-Fi, Thriller. 4. `content` — movies and series (2,500 rows). 5. `review` — user × content × rating × comment. 6. `watchlist` — user × content × status. 7. `tag` and `content_tag` — derived from the source CSV’s `tags` column to power “similar titles”.

Relationships: - `users` 1—N `review` N—1 `content` - `users` N—M `content` (via `watchlist`) - `content` N—1 `platform`, `content` N—1 `genre` - `content` N—M `tag` (via `content_tag`)

The full ER → relation mapping, FDs, and BCNF normalization are in `db_design.pdf`.

5. Technologies

Layer	Choice
Language	Java 17
Framework	Spring Boot 3.2 (Web, Security, Thymeleaf, JDBC)
Database	MySQL 8 in Docker
DB driver	mysql-connector-j (JDBC PreparedStatement only)
Templating	Thymeleaf + Bootstrap 5
Auth	Spring Security <code>formLogin</code> + <code>BCryptPasswordEncoder(12)</code>
Data prep	Python 3 + bcrypt (one-shot to produce data.sql)

No third-party libraries beyond what Project 2 already permitted.

6. User interfaces (≥ 5 required; we ship 8)

1. `/signup` — sign-up form
2. `/login` — login form
3. `/home` — trending titles
4. `/search` — search & filter results
5. `/movies/{id}` — title detail + reviews + similar titles
6. `/watchlist` — user's saved titles
7. `/analytics` — cross-platform analytics dashboard
8. `/admin` — admin-role summary page

Every page's primary content is generated from a SQL query against the `stayanalytics` database; the full query catalog lives in `queries.sql`.